

Kishore Manchikatla

Senior Platform Engineer

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Summary

Platform Engineer with 11+ years of experience building and operating cloud-native infrastructure at scale. At Hexagon, I own the GitOps platform across 10+ environments and 5 AWS regions — from Crossplane-managed AWS infra to a custom Kubernetes operator in Go. I combine deep infrastructure engineering with AI tooling: I built Argus, an open-source AI security agent, and am building a FinOps agent that uses Claude and AWS Cost Explorer MCP to automate cost anomaly investigation. Comfortable working across the full platform stack: Terraform, EKS, Karpenter, ArgoCD, Crossplane, Helm, and Go.

Core Skills

Kubernetes & GitOps: EKS, Karpenter, ArgoCD, Helm, Custom Operators (Go / controller-runtime), Crossplane, IRSA

AWS: EC2, RDS/Aurora, S3, IAM, KMS, ElastiCache, EFS, ECR, WAF, GuardDuty, Security Hub, Secrets Manager, CodeArtifact, CloudWatch

IaC: Terraform, Crossplane (Upbound providers), Helm

Languages: Go, Python, Bash, YAML/Helm templating

Security: Snyk, Semgrep, Trivy, Gitleaks, Bandit, CodeQL, Pod Security Standards, Network Policies, SARIF

Observability: Datadog, Prometheus, Thanos, Grafana

CI/CD: GitLab CI, ArgoCD GitOps, Docker

AI/LLM: Anthropic Claude API, MCP servers, tool-use agent loops, Claude Code

Previous: Oracle WebLogic, MyST automation, on-premises CI/CD

Experience

Hexagon Digital Reality (HxDR) — Platform Engineer

August 2021 – Present | Hyderabad, India

HxDR is a cloud-native SaaS platform for 3D digital reality, running on AWS EKS across **5 regions**, **10+ environments**, and **20+ microservices**.

Kubernetes Operator (Go)

- Designed and maintain a custom Kubernetes operator (Go, controller-runtime) that automates developer environment lifecycle — provisioning, expiry, teardown, and multi-region promotion — reducing per-environment setup from hours to minutes
- Operator integrates with GitLab MR workflows to trigger automated config promotions across environments via custom CRDs

GitOps Platform

- Own the end-to-end GitOps platform built on ArgoCD ApplicationSets — deploys 20+ microservices and 30+ platform operators across all environments using a merge-generator pattern with base + per-cluster overrides

- Architected a versioned platform manifests library for platform tool resources (Karpenter, Datadog, Grafana, Crossplane providers), reducing duplication across clusters and enabling independent versioned releases

AWS Infrastructure (Crossplane + Terraform)

- Manage all AWS infrastructure declaratively via Crossplane: RDS/Aurora, S3, IAM, IRSA, KMS, Redis/ElastiCache, EFS, ECR, WAF, Secrets Manager, CodeArtifact, GuardDuty, Security Hub, AWS Config
- Designed a 10-layer Terraform workspace pattern for foundational EKS infrastructure — layered from networking through security across all AWS accounts (dev/stg/qa/uat/prod)
- Implemented an internal OCI registry (Zot) as a pull-through proxy for provider images, eliminating external registry dependency

Multi-Region & Disaster Recovery

- Onboarded 2 new AWS regions end-to-end — infrastructure, GitOps, and microservice config — from scratch
- Led the Disaster Recovery exercise (regional outage scenario) — achieved all recovery goals within the 40-hour target window (Sep 2025)
- Own and maintain the company-wide Disaster Recovery Plan (DRP)

Security Hardening

- Evaluated and implemented K8s Pod Security Standards (PSS) enforcement and Default-Deny network policy posture across EKS clusters
- Established centralised vulnerability policy management (Snyk) across all microservice repositories with global defaults and per-service overrides
- Enabled GuardDuty, Security Hub, and AWS Config across all accounts via Crossplane; reduced GuardDuty spend by disabling unused regions — saving **~\$14,000/year**

Karpenter & FinOps

- Manage Karpenter NodePool and EC2NodeClass resources across all clusters; tuned spot/on-demand mix, instance families, and consolidation settings to optimise node utilisation
- Reduced dev account monthly cloud spend from **\$8,742 peak** → **~\$3,545** — an **annualised saving of ~\$29,000/year** — through right-sizing, idle resource cleanup, and regional decommissioning

Observability

- Deploy and manage Datadog, Prometheus with Thanos sidecar (multi-cluster metrics), and Grafana with custom dashboards across all clusters via GitOps

Rubicon Red Software India Private Limited — Engineer IV

August 2014 — August 2021 | Hyderabad, India

Rubicon Red is an Oracle Platinum partner delivering enterprise middleware and cloud solutions to global enterprise clients.

AWS Cloud Migration — Vodafone Ireland

- Led migration of on-premises workloads to AWS for Vodafone Ireland, one of Europe's largest telecom providers
- Designed and delivered end-to-end environments in **4–8 hours** vs. weeks or months manually — using EC2, EFS, EC2 Image Builder, ECS Fargate, RDS, Lambda, S3, ECR, ALB, Route 53, CodeBuild,

CodeCommit, Systems Manager

- Built CloudWatch monitoring solutions providing full observability across EC2 instances, ALB, and application logs

Release Automation — Inland Revenue Department, New Zealand

- Automated the provisioning, build, and deployment of **1,300+ releases across 9 environments in 3 months** — saving multiple years of equivalent manual work
- Delivered repeatable, auditable pipelines replacing a previously manual, weeks-long process per environment

WebLogic Platform Engineering

- Oracle WebLogic administration for enterprise clients — domain configuration, cluster management, patching, and performance tuning
 - Led internal adoption of **MyST** (Rubicon Red's WebLogic automation tool) for infrastructure-as-code provisioning and deployment automation across on-premises and cloud environments
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Projects

Argus — AI Security Agent (*Open Source, GitHub*)

Python, FastAPI, Anthropic Claude API, Semgrep, Trivy, Gitleaks, Bandit, CodeQL, OSV-Scanner

- Built a production-grade AI security agent that runs 6 open-source SAST/SCA scanners in parallel and uses Claude to deduplicate findings, chain related vulnerabilities, generate CVSS scores, CWE mappings, and remediation with secure code examples
- **4 usage modes:** CLI, REST API, MCP server (Claude Code integration), CI/CD templates (GitHub Actions, GitLab CI, Azure Pipelines)
- Features: false-positive triage agent, fix generation agent (produces `git apply` -ready diffs), EPSS + CISA KEV enrichment, incremental scan cache, baseline mode, SARIF 2.1.0 + GitLab SAST output, HTML dashboard, Slack/Jira/Teams/GitHub PR/GitLab MR notifications
- 200+ tests; self-scanned with a risk score of 18/100

FinOps Agent (*In Progress*)

Python, Anthropic Claude API, AWS Cost Explorer MCP

- AI agent that uses Claude with the AWS Cost Explorer MCP server to automate cost anomaly investigation — queries spend data, identifies root causes, and generates actionable remediation reports
 - Applied to real HxDR accounts to identify \$29k+/year in savings
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Education

B.Tech, Computer Science & Engineering (CSE) — Jawaharlal Nehru Technological University, Hyderabad | 2010 – 2014

Notable

- Attended **KubeCon India 2025** — presented internal takeaways on Kube Resource Orchestrator (KRO) to the platform team

- Postmortem owner: P1 incident (accidental deletion of a cluster-wide ArgoCD resource — caused node loss across all environments, Jun 2024)
- ~1,000 commits across HxDR platform repos since 2021